

	B200			GH200			MI300X
	Triton	cuTile	TileLang	Triton	cuTile	TileLang	Triton
matmul_fp32_fp16_fp8 · fp32	TMA TMEM SMEM	TMA TMEM SMEM	cp.async SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS
matmul_fp32_fp16_fp8 · fp16	TMA TMEM SMEM spill	TMA TMEM SMEM	cp.async TMEM SMEM spill	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS cvt
matmul_fp32_fp16_fp8 · fp8_e4m3fn	TMA TMEM SMEM	TMA TMEM SMEM	cp.async TMEM SMEM	TMA SMEM	TMA SMEM spill	cp.async SMEM	unsupported
matmul_int8 · int8	TMA SMEM	TMA TMEM SMEM	cp.async+LDG TMEM SMEM	TMA SMEM	TMA SMEM	cp.async+LDG SMEM	desc → ptr LDS
batched_matmul · fp32	TMA TMEM SMEM	TMA TMEM SMEM	cp.async SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS cvt
streamk_matmul · fp16	TMA TMEM SMEM spill atom.g	TMA TMEM SMEM atom.g	cp.async TMEM SMEM spill atom.g	TMA SMEM atom.g	TMA SMEM atom.g	cp.async SMEM atom.g	desc → ptr LDS cvt atomic
flash_attention · fp16	TMA TMEM SMEM	TMA TMEM SMEM	LDG TMEM SMEM	TMA SMEM	TMA SMEM	LDG SMEM	desc → ptr LDS spill cvt
block_sparse_attention · fp16	TMA+LDG SMEM spill	TMA+LDG TMEM SMEM atom.g	LDG TMEM SMEM	TMA+LDG SMEM spill	TMA+LDG SMEM spill	LDG SMEM	desc → ptr LDS spill cvt
linear_self_attention · fp32	TMA+LDG TMEM SMEM	TMA+LDG TMEM SMEM spill atom.g	LDG SMEM	TMA+LDG SMEM	TMA+LDG SMEM spill	LDG SMEM	desc → ptr LDS cvt
1d_conv · fp16	LDG TMEM SMEM	LDG SMEM spill	LDG TMEM SMEM	LDG SMEM	LDG SMEM	LDG SMEM	pointer LDS
3d_conv · fp32	cp.async TMEM SMEM	LDG TMEM SMEM	LDG SMEM spill	cp.async SMEM spill	cp.async SMEM spill	LDG SMEM spill	pointer LDS
destindex · int8	LDG –	LDG SMEM	LDG –	LDG –	LDG SMEM	LDG –	pointer –
histogramming · int32	LDG SMEM atom.g	TMA+LDG SMEM atom.g	LDG SMEM atom.s	LDG SMEM atom.g	TMA SMEM atom.g	LDG SMEM atom.s	pointer LDS cvt atomic
2d_max_pooling · fp32	LDG SMEM	LDG SMEM	LDG –	LDG SMEM	LDG SMEM	LDG –	pointer cvt
vector_add · fp32	LDG –	LDG –	LDG –	LDG –	LDG –	LDG –	pointer –

Line 1: operand load path, union over the kernels of the run (TMA = TMA loads; cp.async; LDG = ordinary global loads; desc → ptr = TensorDescriptor lowered to pointer loads; pointer = pointer loads). Line 2: SMEM / LDS / TMEM use, spill (local or scratch), atom.g / atom.s (global / shared atomics at scale), ldsm (LDSTM/STSM layout operations), cvt (convert_layout).

tcgen05
legacy MMA (HMMA/IMMA)
no matrix instr.
unsupported
WGMMMA
MFMA
static SASS/ISA only